Understanding vector graphics and bitmaps

There are two main types of computer graphics. They are vector graphics and bitmaps.

Vector graphics

Vector graphics are made of **lines and curves**. They are generated from **mathematical descriptions**. These graphics are ideal for **logos and illustrations**. As they are **resolution-independent graphics**, they can be scaled to any size, or printed and displayed at any resolution, without losing detail and quality.

Bitmaps

Bitmaps are also known as **raster images**. They are composed of small squares called **pixels**. Each pixel is mapped to a location in an image and has numerical colour values.

As Bitmaps reproduce color gradations well, they are excellent for photographs and digital paintings.

Bitmaps are **resolution-dependent images** which means that they represent a fixed number of pixels. They look good at their actual size, but they appear jagged or lose image quality when scaled, or when displayed or printed at a resolution higher than their original resolution.

By using CorelDRAW you can create vector graphics. You can also import bitmaps(raster images) such as JPEG and TIFF files into CorelDRAW and add them to your drawings.

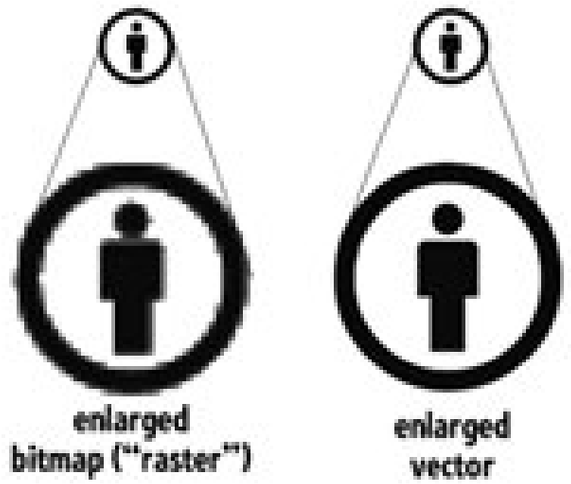


Fig. 3.2 Bitmap image and vector image

If you enlarge a bitmap in your page layout package, you may lose resolution and it may appear jagged. If you shrink a bitmap in your page layout package, the result will look good.

CorelDRAW Terms

Before getting started with CorelDRAW you should be familiar with the following terms.

Term	Description
Object	An object is an element in a drawing such as an image, shape, line,text, curve, symbol, or layer.
Drawing	Drawing is the work you create in CorelDRAW, for example, custom artwork, logos, posters, and newsletters.
Vector graphic	Vector graphic is an image generated from mathematical descriptions that determine the position, length, and direction in which lines are drawn.

XII CompTech_English version CHAPTER 03.indd 86 01-03-2020 13:05:05